

MS-Excel

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Microsoft Excel is a proprietary commercial spreadsheet application. It is an electronic spreadsheet program that can be used for storing, organizing and manipulating data. It features calculation, graphing tools, pivot tables and a macro programming language. It is a very widely applied spreadsheet. It is compatible with multiple OS like Window, macOS, Android and iOS.

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How to Open Microsoft Excel?

Running Excel is not different from running any other Windows program. If you are running Windows with a GUI like (Windows XP, Vista, and 7) follow the following steps:

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- Click on start menu
- Point to all program
- Point to Microsoft Excel
- Click on Microsoft Excel

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Alternatively, you can also open it from the start menu if it has been added there.

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You can also open it from the desktop shortcut if you have created one.

Understanding the Ribbon

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The ribbon provides shortcuts to commands in Excel. A command is an action that the user performs. An example of a command is creating a new document, printing a document, etc.

Ribbon components explained

Ribbon start button :-

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It is used to access commands i.e. creating new documents, saving existing work, printing etc.

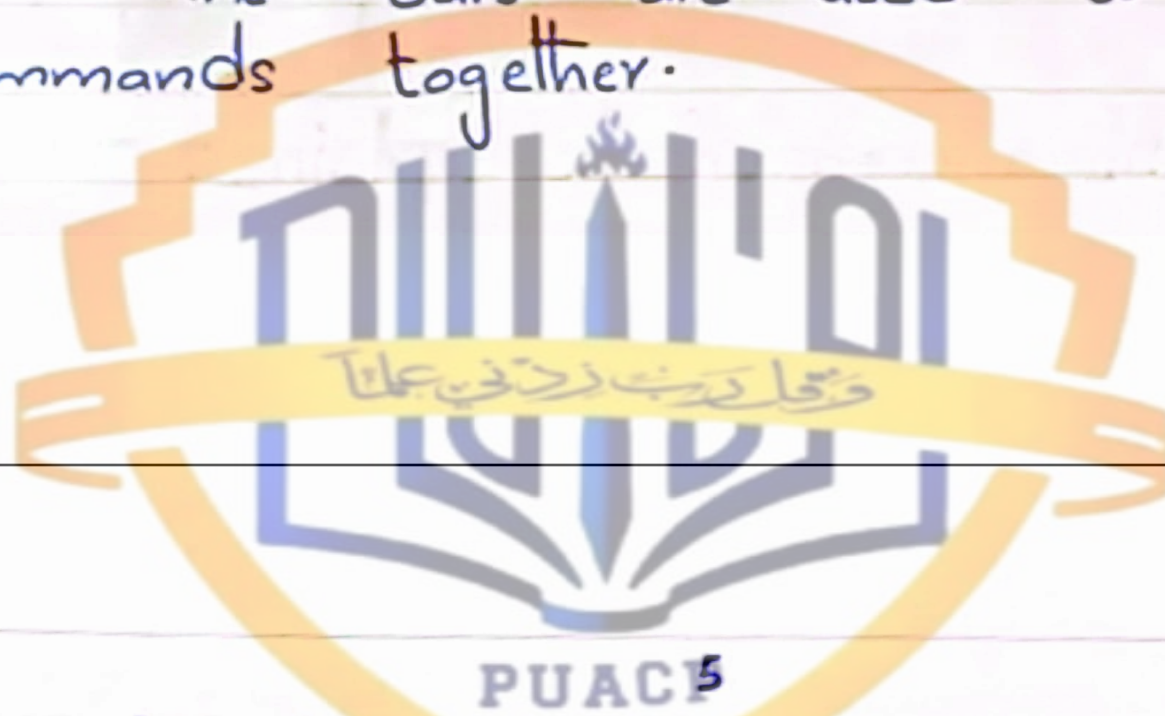
Ribbon tabs :-

The tabs are used to group similar commands together.

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Ribbon bar :-

The bars are used to group similar commands together.



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Understanding the worksheet (Rows and Columns, Sheets, Workbooks)

A worksheet is a collection of rows and columns. When a row and a column meet, they form a cell. Cells are used to record data. Each cell is uniquely identified using a cell address. Columns are usually labelled with letters while rows are usually numbers.

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A workbook is a collection of worksheets. By default, a workbook has three cells in Excel. You can delete or add more sheets to suit your

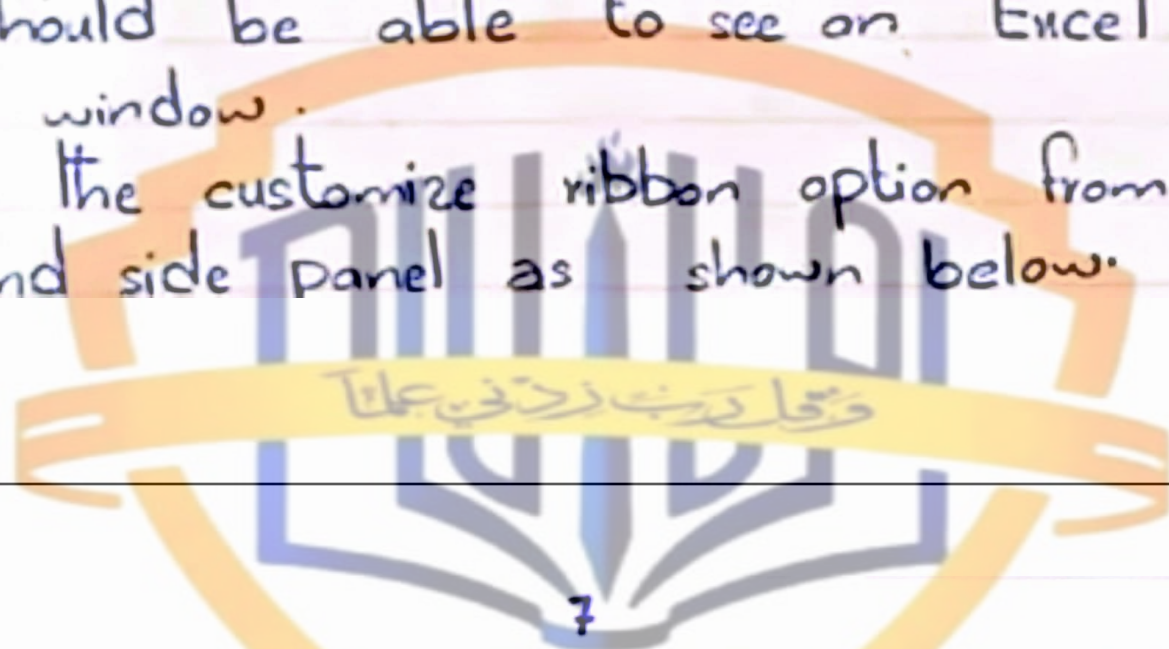
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requirements. By default, the sheets are named Sheet 1, Sheet 2 and so on and so forth. You can rename the sheet names to more meaningful names i.e. Daily Expenses, Monthly Budget, etc.

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Customization of ribbon

- Click on the ribbon start button
- Select options from the drop down menu. You should be able to see on Excel Options dialog window.
- Select the customize ribbon option from the left-hand side panel as shown below.



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- On your right-hand side, remove the check mark from the tabs that you do not wish to see on the ribbon. For this example, we have removed Page Layout, Review, and View tab.
- Click on the "OK" button when you are done.

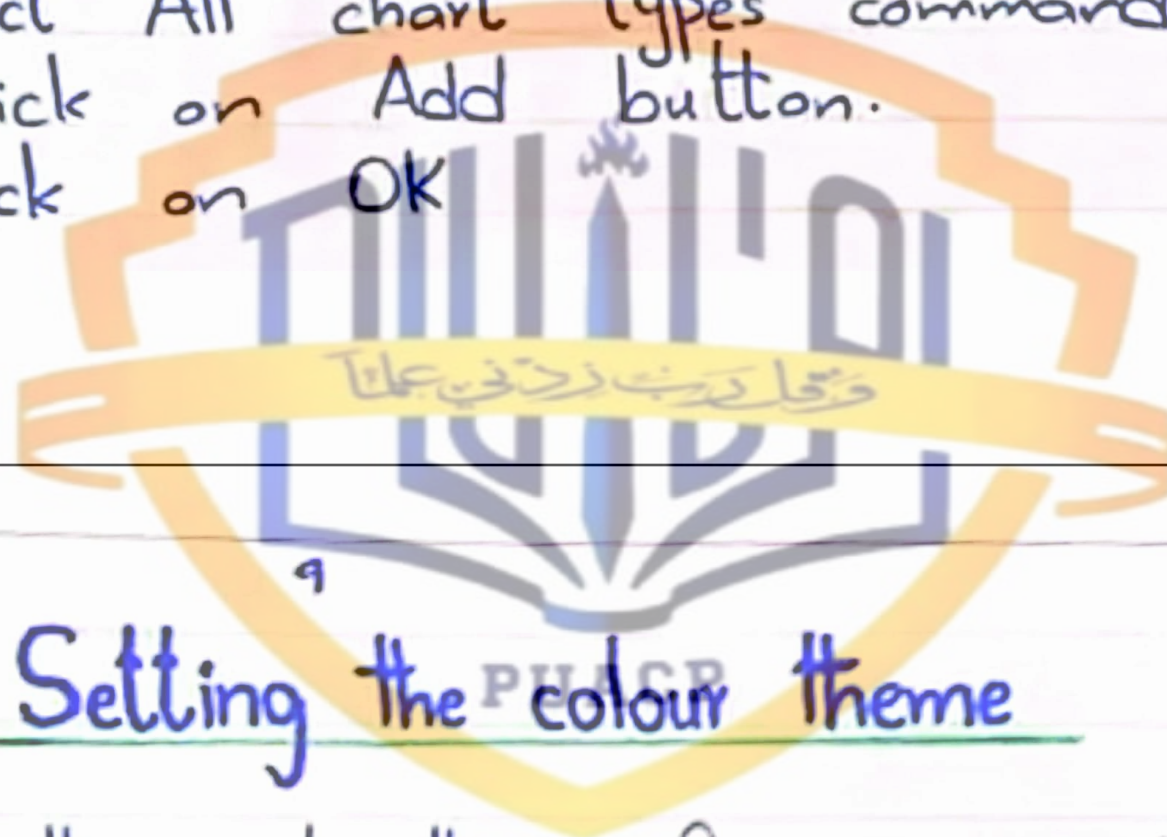
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Adding custom tabs to the ribbon

- 1- Right click on the ribbon and select Customize the Ribbon. The dialogue window shown above will appear.
- 2- Click on new tab button as illustrated in the animated image below.

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- 3- Select the newly created tab
- 4- Click on Rename button
- 5- Give it a name of Guru99 **BS Notes Bank**
- 6- Select the New Group (Custom) under Guru99 tab as shown in the image below.
- 7- Click on Rename button and give it a name of My Commands
- 8- Let's now add commands to my ribbon bar.
- 9- The commands are listed on the middle panel.
- 10- Select All chart types command and click on Add button.
- 11- Click on OK



Setting the colour theme

To set the color-theme for your Excel sheet you have to go to Excel ribbon, and click on a File à Option command. It will open a window where you have to follow the steps.

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- The general tab on the left-hand panel will be selected by default.
- Look for colour scheme under General options for working with Excel.
- Click on the colour scheme drop-down list and select the desired colour
- Click on OK button.

Setting for formulas

This option allows you to define how Excel behaves when you are working with formulas. You can use it to set options i.e. autocomplete when entering formulas, change the cell referencing style and use numbers for both columns and rows and other options. **BS Notes Bank**

Proofing settings

This option manipulates the entered text entered into excel. It allows setting options such as the dictionary language that should

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be used when checking for wrong spellings, suggestions from the dictionary, etc.

Save settings

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This option allows you to define the default file format when saving files, enable auto recovery in case your computer goes off before you could save your work, etc. You can use this option from the Options dialogue window under save tab from the left hand side panel.

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Important Excel shortcuts

- Ctrl + P used to open the print dialogue window. **BS Notes Bank**
- Ctrl + N creates a new workbook.
- Ctrl + S saves the current workbook.
- Ctrl + C copy contents of current select.
- Ctrl + V paste data from the clipboard.
- SHIFT + F3 displays the function insert dialog window
- SHIFT + F11 Creates a new worksheet
- F2 Check formula and cell range covered.

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II FUNCTION IN EXCEL

Introduction: IF function is one of the most important functions in excel. IF function evaluates a boolean/logical expressions and returns one value if the expression evaluates to TRUE and another value if the expression evaluates to FALSE.

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Definition:-

According to Microsoft Excel, IF function is defined as a formula which "checks whether a condition is met, returns one value if true and

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another value if false"

Syntax:

Syntax of IF function in Excel is as follows:

=IF (logic_test,
[value_if_true]
[value_if_false])

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- 'logic_test' (required argument) - Refers to the boolean expression or logical expression that needs to be evaluated.

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- 'value_if_true' (optional argument) - Refers to the value that will be returned by the IF function if the 'logic_test' evaluates to TRUE.
- 'value_if_false' (optional argument) - Refers to the value that will be returned by the IF function if the 'logic_test' evaluates to FALSE.

Important Characteristics of IF Function in Excel

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- To use the IF function, you need to provide the 'logic_test' or conditional statement mandatorily.

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- The arguments 'value_if_true' and 'value_if_false' are optional, but you need to provide at least one of them.

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- The result of the IF statement can only be any one of the two given values (either it will be 'value_if_true' or 'value_if_false'). Both values cannot be returned at the same time.

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- IF function Throws a '#Name?' error if the 'logic_test' or boolean expression you are trying to evaluate is invalid.

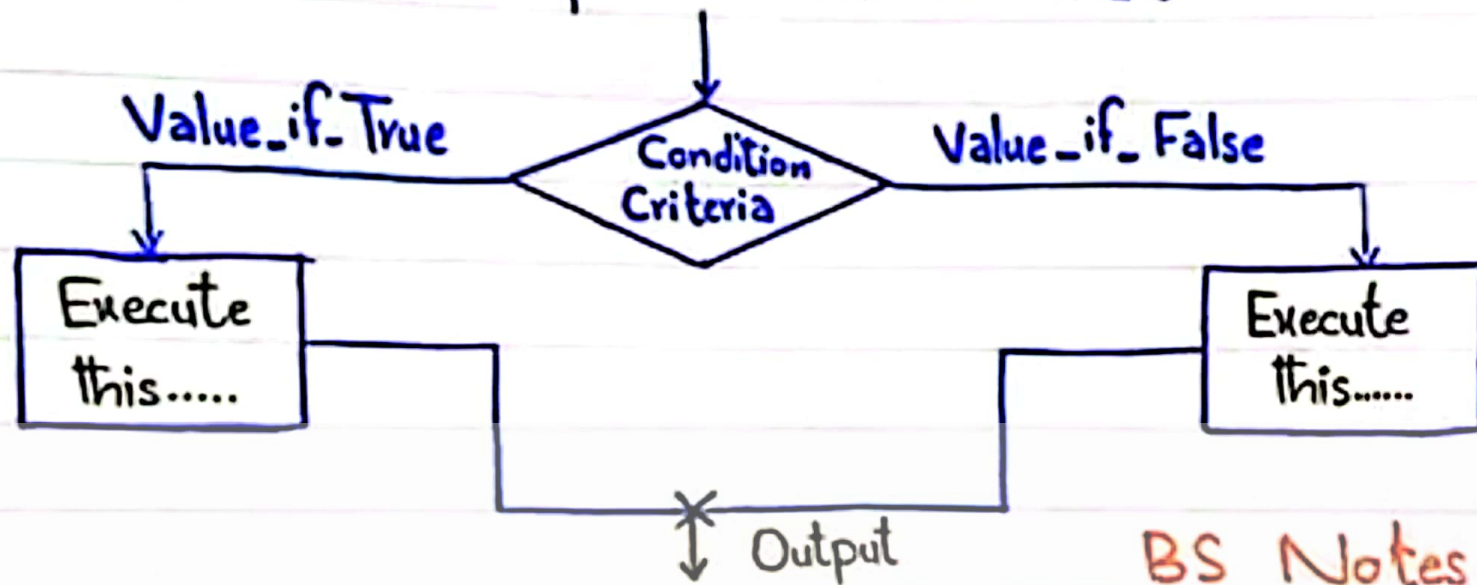
- Nesting of IF statements is possible, but Excel only allows this to 64 levels. Nesting of IF statement means using one if statement within another.

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General Flow Chart Structure of Excel IF Function

- As discussed above, the IF function works on the concept of a flow chart. We can draw a flow chart of generic IF function like below:

Simple IF Function Flow chart



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The above flow chart shows that there is only one condition or criteria, while the two outcomes are based on condition satisfaction. If the condition is satisfied (evaluated as TRUE), the function returns the value from the left box. If the condition is not satisfied (evaluated as false), the function returns the output from the other side.

Logical Operators used in IF Function

The Excel IF formula typically uses logical operators to compare the values based on the given condition. When evaluating a test using the IF functions, we can use any of the below logical operators:

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Operator	Description	Example
=	Equal to	A1=B1
<	Less Than	A1
<=	Less Than or equal to	A1<=B1
>	Greater Than	A1>B1
>=	Greater Than or equal to	A1>=B1
<>	Not equal to	A1<>B1

How to use the IF Function in Excel?

To use the IF function in our Excel sheet, we must perform the following step:

- First, we must enter or type the equal sign (=) to start the function name.

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- Next, we must type the function name, followed by the starting bracket, i.e., =IF(.

- In the next step, we need to give the function the first argument (logical_test).

We must specify the condition using operators' signs for logical comparison.

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- After that, we need to give optional arguments, returning a value as an output based on the TRUE or FALSE.

Excel IF Function Example

Basic IF Function application for empty/non-empty cell:

Everyone knows that life is not possible without oxygen. Suppose we have the following excel

sheet as an example data set where column A contains the list of some planets and column B contains data about the availability of oxygen for these planets. **BS Notes Bank**

	F9	(o	fx		
	A	B	C	D	E
1	Planet Names	Oxygen Availability	Possible of Life		
2	Jupiter	No	????	←	
3	Venus	No	????		
4	Mars	No	????		
5	Earth	Yes	????		
6					

- We only need to use the Excel IF Function to find the planets where life is possible. We can use the oxygen availability criteria in the IF function to get the desired result.

Let us now put the IF formula in our resulting column C (cells C2 to C5) and find the names of the planets with the potential for life among the planets listed in our example sheet: **BS Notes Bank**

- First, we find the possibility of life on the first planet (i.e., Jupiter). So, we select the respective resultant cell C2 and start the formula by entering an equal sign.

After that, we enter the IF function name and the starting bracket, as shown below:

	SUM	(X ✓ ✗)	=IF(		
	A	B	C	D	E
1	Planet Names	Oxygen Availa- bility	Possible of Life		
2	Jupiter	No	=IF(1	←	
3	Venus	No	? IF(logical-test, [value-if-true], [value-if-false])		
4	Mars	No	????		
5	Earth	Yes	????		
6					

- Next, we apply the condition for the first planet. Since we are checking for oxygen availability as the life factor, we apply the condition (B2="YES") as B2 has data for oxygen availability. It will look like this: BS Notes Bank

	SUM	(X ✓ ✗)	=IF(B2="YES"		
	A	B	C	D	E
1	Planet Names	Oxygen Avail ability	Possible of life		
2	Jupiter	No	=IF(B2="YES",		
3	Venus	No	? IF(logical-test, [Value-if-true], [value-if-false])		
4	Mars	No	????		
5	Earth	Yes	????		

- Lastly, we give values to be returned based on the satisfaction and dissatisfaction of the condition. If the condition matches, we want the IF function to return: 'Life is possible here'. If the condition does not match, we want the IF function to return: 'Life is not possible here'. We use the Comma (,) to separate the arguments. So, the entire formula looks like this:

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SUM		(X ✓ formula) =IF(B2="YES" Life is possible here, "Life is not possible here")		
A		B	C	D
1	Planet Names	Oxygen availability	Possible of life	
2	Jupiter	No	=IF(B2="Yes", "Life is possible here", "Life is not possible here")	
3	Venus	No	? IF(logical-test, [value-if-true], [value-if-false])	
4	Mars	No	????	
5	Earth	Yes	????	
6				

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SUM		(X ✓ fx =IF(B2="YES"Life is possible here"Life is not possible here"			
	A	B	C	D	E
1	Planet Names	Oxygen availability	Possible of life		
2	Jupiter	No	=IF(B2="Yes", "Life is possible here", "Life is not possible here")		
3	Venus	No	?IF(logical-test, [value-if-true], [value-if-false])		
4	Mars	No	????		
5	Earth	Yes	????		
6					

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After that, we type the closing bracket and press the Enter key to get the corresponding result. This provides the result that life is not possible on Jupiter.

C2		(fx =IF(B2="YES"(Life is possible here"Life is not possible here")			
	A	B	C	D	E
1	Planet Names	Oxygen Availability	Possible of life		
2	Jupiter	No	Life is not possible here		
3	Venus	No	????		
4	Mars	No	????		
5	Earth	Yes	????		
6					

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- After finding the result for the first planet, we can apply the IF formula in the remaining resultant cells accordingly. However, it will be easier to copy-paste the formula from C2 to other cells below. Also,

we can drag the formula using the Fill Handle. This will help us perform the task quickly.

C2		fn =IF(B2="YES"Life is possible here"Life is not possible here"			
	A	B	C	D	E
1	Planet Names	Oxygen Availability	Possible of life		
2	Jupiter	No	Life is not possible here		
3	Venus	No	Life is not possible here	←	
4	Mars	No	Life is not possible here		
5	Earth	Yes	Life is possible here		
6					

Nested IF Function

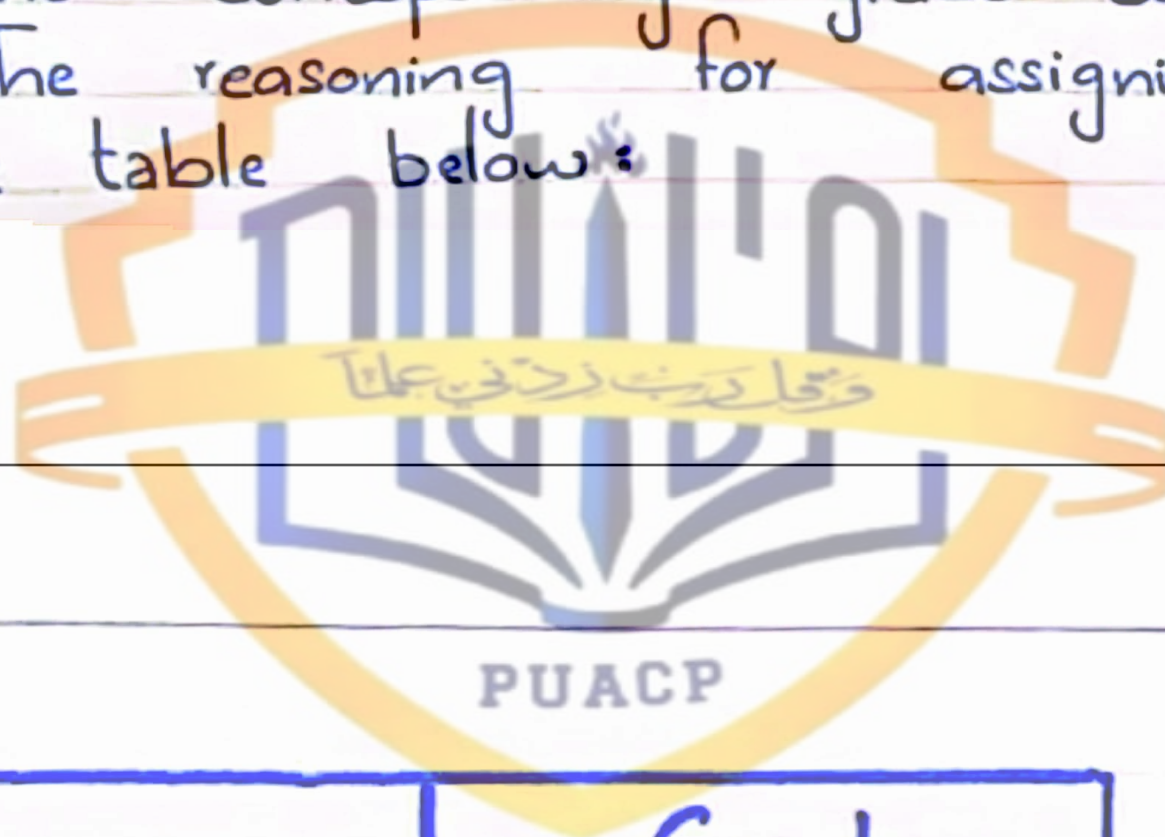
The IF function in Excel can be nested. The Nested IF function is a combination of multiple IF functions. A 'nested IF' contains at least one IF function inside another to evaluate more than one condition at once and return the output accordingly. In Excel, up to 64 IF functions can be nested in a formula. But, we must double-check to ensure that each IF condition is correctly nested inside the other.

The following example shows the use of nested IF where IF functions are used (nested) inside another.

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`=IF(C5<60,"F",IF(C5<71,"D",IF(C5<85,"C",IF(C5<95,"B","A"))))`

The condition in the IF function is used to find the corresponding grade based on the scores. The reasoning for assigning grades is in the table below:



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Score	Grade
0-60	F
61-70	D
71-84	C
85-94	B
95-100	A

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Score	Grade
0-60	F
61-70	D
71-84	C
85-94	B
95-100	A

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We usually move in a nested IF formula in one direction, whether from high to low or low to high. In our example, we go from low to high. We start by testing to find grades for scores below 60, then move on to the next IF function and test grades for scores below 70, and so on.

